Testing Results Report

DOST Meeting Management: Handover Validation

Date: 2026-10-07

This report records the automated validation of the DOST Meeting Management system for project handover, and a local smoke test of the running system. It replaces the report of 3 June 2026, which covered 9 server test suites and predates most of the delivered code.

Summary

Area	Status	Evidence
Linting	**Passed**	ESLint reported no errors in the client and server workspaces.
Server tests without a database	**Passed**	35 suites and 453 tests passed; 144 database-backed tests were skipped.
Server tests with PostgreSQL	**Passed**	All 43 suites and 597 tests passed, including the database-backed tests.
Client tests	**Passed**	62 tests passed.
AI service tests	**Passed**	16 of 20 tests passed; 4 were skipped because they need PyTorch.
Client production build	**Passed**	Vite built the production bundle (2,796 modules).
Local smoke test (API level)	**Passed**	38 of 38 checks over HTTP on an isolated local stack, with real Whisper and Ollama processing.
Browser walkthrough	**Not run for this report**	Manual procedure; see "Validation not covered by this report".
Smoke test on a deployed host	**Not run for this report**	Manual procedure; see "Validation not covered by this report".

Test Environment

Item	Value
Code version	Commit `b075593` on branch `jhowliu/deployment-guide-managed-options`, based on `main` at `f9dd2ab`
Host	macOS 15.7.4
Node.js	22.16.0 with npm 11.4.1; Jest 30.2.0 for server tests, the Node test runner for client tests
Database	PostgreSQL 16.15 (`postgres:16-alpine`) in a disposable container with the `pgcrypto` and `uuid-ossp` extensions
Python	3.13.0 in a virtual environment with `ai-service/requirements.fixture.txt` installed (no PyTorch)
Mocked services	Redis and object storage are replaced by test doubles in the automated server tests

The database-backed tests truncate tables, so they ran against the disposable database rather than a development database. Every migration in `server/db/migrations` was applied to the empty database before the

tests ran.

Commands and Observed Results

Linting

```
npm run lint
```

```
> @meeting-mgnt/client@0.0.0 lint
> eslint .

> @meeting-mgnt/server@0.1.0 lint
> eslint .
```

ESLint printed no warnings or errors, and the command exited with status 0.

Server tests without a database

```
npm run test --workspace=server
```

```
Test Suites: 8 skipped, 35 passed, 35 of 43 total
Tests:       144 skipped, 453 passed, 597 total
Time:        3.361 s
```

Server tests with PostgreSQL

```
npm run test:db --workspace=server
```

```
Test Suites: 43 passed, 43 total
Tests:       597 passed, 597 total
Time:        21.699 s
```

Client tests

```
npm run test --workspace=client
```

```
# tests 62
# suites 3
# pass 62
# fail 0
# skipped 0
```

AI service tests

```
cd ai-service
python -m unittest discover -s tests -t .
```

```
Ran 20 tests in 0.299s
OK (skipped=4)
```

The four skipped tests check the CPU and GPU compute profiles and are skipped when PyTorch is not installed.

Client production build

```
VITE_API_BASE_URL=/ npm run build --workspace=client
```

```
✓ 2796 modules transformed.
✓ built in 655ms
```

Local Smoke Test

A smoke test drove the running system over HTTP with a script, on an isolated local stack, on 7 October 2026. It used the same code as the automated tests (commit `b075593`) and real Whisper and Ollama processing. It did not use a browser, and it did not run behind Caddy or under systemd.

Item	Value
Host	Apple M2 Pro, 32 GiB RAM, macOS 15.7.4, Docker 28.0.4
API and worker	Node.js 22.16.0, started directly (`node index.js`, `node worker.js`) with `AI_PIPELINE_ENABLED=true`
Data services	PostgreSQL 16 (`postgres:16-alpine`), Redis 7 (`redis:7-alpine`, `noeviction`), Silo (`pgsty/silo:RELEASE.2026-09-16T00-00-00Z`), each in a disposable container
Email	SMTP pointed at a local Mailpit container, so messages were captured and none left the host
AI worker	`ai-service` container image with the current `app/` code; Whisper `small` on CPU
LLM	Ollama 0.35.0 running natively on macOS (Apple GPU) with `llama3`
Recording	`ai-service/test-fixtures/input/power_system_meeting_3min_final.mp3` (3 min 4 s, 4.2 MB)
Data	A new empty database; the first administrator was created with `npm run create-admin`, and all other data by the script

Results

All 38 checks passed.

#	Area	Check	Expected	Result
1	Health and accounts	GET /health/live	200	200
2	Health and accounts	GET /health/ready (PostgreSQL, Redis, object storage)	200	200 (postgres:ok, redis:ok, minio:ok)
3	Health and accounts	Unauthenticated GET /api/meetings	401	401
4	Health and accounts	Administrator sign-in (POST /api/auth/login)	200	200

#	Area	Check	Expected	Result
5	Health and accounts	Self-registration closed (POST /api/auth/register)	403	403
6	Health and accounts	Administrator creates a staff account (POST /api/users)	201	201
7	Health and accounts	Administrator creates a second staff account	201	201
8	Health and accounts	Staff sign-in	200	200
9	Health and accounts	Staff cannot create accounts (POST /api/users)	403	403
10	Health and accounts	Dashboard (GET /api/dashboard)	200	200
11	Scheduling and email	Create meeting request (POST /api/meeting-requests)	201	201
12	Scheduling and email	Add time options	200	200
13	Scheduling and email	Add an external invitee	200	200
14	Scheduling and email	Send invitations (POST .../send-invitations)	200	200
15	Scheduling and email	Invitation email delivered to the invitee (Mailpit)	one message	"Meeting invitation: Smoke Test Review"
16	Scheduling and email	Invitation email carries a response link	token in link	found
17	Scheduling and email	Open the invitation link without signing in (GET /api/public/invitations/:token)	200	200
18	Scheduling and email	Invitee submits availability through the link	200	200 (2 options)
19	Scheduling and email	Request moves to ready_to_confirm after all invitees respond	ready_to_confirm	ready_to_confirm
20	Scheduling and email	Confirm the meeting (POST .../confirm)	2xx	201
21	Scheduling and email	Confirmed request created a meeting	meeting id	present
22	Scheduling and email	Confirmation email delivered	more than one message	3 messages in total
23	Meetings and files	Meeting detail (GET /api/meetings/:id)	200	200
24	Meetings and files	Non-participant cannot open the meeting	403 or 404	403

#	Area	Check	Expected	Result
25	Meetings and files	Administrator sees the meeting in the meeting list	listed	listed
26	Meetings and files	Upload a meeting file	201	201
27	Meetings and files	Download through the short-lived link matches the upload	same bytes	same bytes
28	Meetings and files	Stream file content (GET /api/files/:id/content) matches the upload	same bytes	same bytes
29	Meetings and files	Non-participant cannot read the file	403	403
30	Recording workflow	Recording upload without consent is rejected	400	400
31	Recording workflow	Upload a recording with consent	2xx	201 (4.2 MB)
32	Recording workflow	Upload alone does not start processing	no transcript	no transcript
33	Recording workflow	Start processing (POST .../recording/generate)	2xx	202
34	Recording workflow	Transcription completes (Whisper small, CPU)	completed	completed
35	Recording workflow	Analysis completes and waits for review (Ollama, llama3)	ready_for_review	ready_for_review (see processing times below)
36	Recording workflow	Status updates arrive over SSE	events received	snapshot, status and settled events
37	Recording workflow	Analysis returns a summary for review	non-empty summary	summary of the feeder 7 voltage drop discussion (6 proposed action items)
38	Recording workflow	Apply the reviewed analysis to the minutes	2xx	200

Processing times

From the AI worker log, for the 3-minute recording: transcription took 105.7 s and analysis took 30.6 s. The analysis time is not a CPU-only figure, because Ollama used the Apple GPU. The script's own wall-clock time for this step is longer and is not a processing time: the AI worker was restarted once during the run (see "Findings"), and it picked up the queued job after the restart.

Coverage Against the Technical Specification

Section 9 of the Technical Specification lists the validation scope. The evidence for each area is:

Scope	Evidence	Not covered
Functional testing	Controller and service tests for sign-in, dashboard, meetings, scheduling requests, files, recordings, minutes and action items; client tests for the dashboard, my meetings, action items, recording progress and form validation; the local smoke test runs these flows through the API	No browser end-to-end tests
API testing	Authentication, authorization (403), validation errors (400), pagination and status transitions	Most routes are tested at controller level; only a few run over HTTP (`app.test.js`)
Database testing	Migrations applied to an empty database; schema constraints and relationships (`meeting-transcript-schema.test.js`, `recording-workflow-schema.test.js` and the controller tests)	None identified
File testing	Upload, download link, content stream, delete, size limit and unauthorized access (`file.test.js`, `upload-middleware.test.js`)	Object storage is mocked in the automated tests; the smoke test used Silo
Security testing	Authentication required (401), meeting-scoped permissions, administrator-only actions, rate limiting (429)	No penetration testing
Deployment smoke testing	Local API-level smoke test: health checks, sign-in, scheduling with email, files, and the recording workflow with Whisper and Ollama (38 checks)	Not repeated on a deployed host behind Caddy with TLS and systemd

Validation Not Covered by This Report

- **Browser walkthrough.** The smoke test drove the API directly; the web interface was not exercised. The manual steps are in the Deployment Guide, Step 8.
- **Smoke test on a deployed host.** The smoke test ran the API and workers directly on a workstation, without Caddy, TLS or systemd. Repeat it on the target host after deployment (Deployment Guide, Section 9.9).
- **Load testing.** No load test has been run, and no capacity figure is claimed.

Changes Since the Previous Report

- Server tests grew from 9 suites and 113 tests to 43 suites and 597 tests.
- Client tests (62) and AI service tests (20) are now reported.
- New tests cover the upload size limit and the file content stream (`GET /api/files/:id/content`).

Findings and Known Gaps

- **Upload size limit.** A file must be smaller than 50 MB. A file of exactly 50 MB (52,428,800 bytes) is rejected with HTTP 413, because the upload library stops a file once it reaches the limit.
- **Continuous integration.** CI runs linting, the server tests without a database, the client tests and the client build. The database-backed server tests and the AI service tests run only manually.
- **Mocked services.** Redis and object storage are mocked in the automated tests; the local smoke test exercised the real services.
- **Database connection loss stops the API and workflow worker.** While the smoke environment was being reset, PostgreSQL terminated the open connections ("terminating connection due to administrator command"), and both the API and the workflow worker exited. The PostgreSQL connection pool (`server/db/postgres.js`) has no `error` handler, so an error on an idle connection ends the process. The

systemd units restart a failed service (`Restart=on-failure`), but requests fail until then. A database restart or a managed-database failover (Deployment Guide, Options 2 and 3) would have the same effect.

- **AI worker and a cleared Redis.** Clearing Redis during the reset also removed the AI worker's consumer group, and the worker exited. After a restart it recreated the group and processed the job that had been queued in the meantime.

Conclusion

All automated checks passed for commit `b075593`: linting, 597 server tests with PostgreSQL, 62 client tests, 16 AI service tests (4 skipped without PyTorch) and the client production build. The local smoke test passed all 38 checks, including a 3-minute recording processed by Whisper and Ollama. A browser walkthrough and a smoke test on a deployed host remain to be done.